

lib/ Hierarchy & Roles

Architecture and Class Responsibilities for ineTeam

1. Core Layer

core/

utils/helpers.dart

Global utility functions for formatting, icons, and color mapping. **UTILITY**

2. Data Layer

data/

models/*.dart

Data classes (MatchModel, UserModel) for serialization and state handling. **POJO / MODEL**

repositories/*.dart

High-level data access layer. Mediates between providers and services. **REPOSITORY**

services/*.dart

Direct interaction with Firebase (Auth, Firestore, Storage). **INFRASTRUCTURE**

3. Feature Layer (State Management)

features/

auth/auth_provider.dart

Manages user authentication state and session persistence. **PROVIDER**

matches/match_provider.dart

State for match discovery, creation, and participating lists. **PROVIDER**

matches/matchmaking_service.dart

Logic for team balancing and matchmaking algorithms. **SERVICE LOGIC**

4. Presentation Layer (UI)

presentation/

screens/*.dart

Primary views (Home, Profile, MatchDetail) tied to providers. **SCREEN / VIEW**

widgets/match_card.dart

Unified, reusable match component with status indicators. **WIDGET**

widgets/team_balance_view.dart

Visual representation of team slots and distributions. **WIDGET**